Summary in Graph

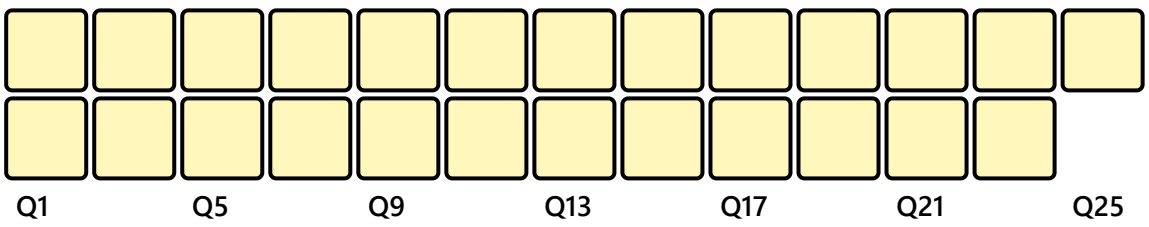
Exam Summary (GO Classes CS Test Series 2026 | CO and Architecture | Subject Wise Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	25 (10 + 15)
Total Marks:	40 (10 + 30)
Exam Duration:	72 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (92) out of 916 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 CO & Architecture

A computer memory system has a $64\ \mathrm{KB}$ byte-addressable main memory with 16-bit addresses. This same system has a one-level cache memory that can hold 16 blocks of data, where each block contains 16 bytes. Assuming this is a direct-mapped cache, to which cache block will main memory address 39553_{10} map?

A. 0_{10}
B. 1_{10}

C. 8_{10}
D. 9_{10}

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

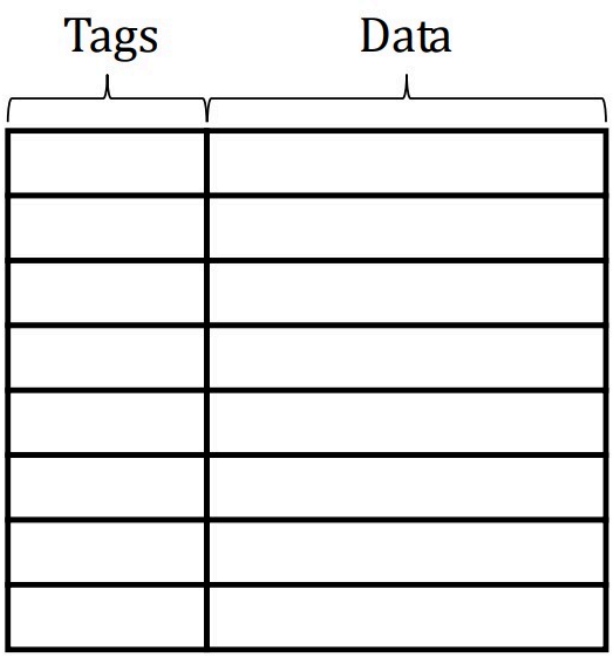
Q #2 Numerical Type | Award: 1 | Penalty: 0 | CO & Architecture

Suppose that the level 1 data cache has a hit rate of 40% on your application, an access time of a single cycle, and a miss penalty to memory of forty cycles. What is the average memory access time (in cycles)?

Your Answer: | Correct Answer: 25 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Numerical Type | Award: 1 | Penalty: 0 | CO & Architecture

A processor has an 8-line level 1 directly mapped data cache as shown below. Suppose that cache lines are 32 bytes (256 bits) and memory addresses are 16 bits, with byte-addressable memory.



How many bits in total are needed to implement the level 1 data cache ?

Your Answer: | Correct Answer: 2112 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

A fully-associative cache is like a

A. direct-mapped cache with only one set
B. direct-mapped cache with only one line per set
C. set-associative cache with only one line per set
D. set-associative cache with only one set

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Evaluate the following statements and select the appropriate answer given from the choices below.

I. Von Neumann Architecture shares common memory for Data and Instructions

II. Harvard Architecture has separate physical memories for Data and Instructions

A. Only I is true
B. Only II is true
C. Both I and II are true
D. None of them is true

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

If the Most Significant Byte (MSB) is stored first while ordering byte values for storing data in memory, it is called as

A. Big-endian
B. Little-endian
C. Big- and Little-endian
D. None of them

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Instead of 2 memories, one to store instructions and another one to store data, if we have only one memory then which type of hazard we could get?

A. Data hazard
B. Control hazard
C. Structural hazard
D. No hazard

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

If the initial register contents of R0, R1 and R2 were

$$\mathrm{R0} = 0 \times 00000000$$
$$\mathrm{R1} = 0 \times 02040608$$
$$\mathrm{R2} = 0 \times 10305070$$

Assume R0 is the result register, after one of the operations below was performed on R1 and R2, which has been modified to R0 $= 0 \times 12345678$
What was the operation performed on the contents of R2 and R1?

A. Bitwise AND
B. Bitwise OR
C. Bitwise XOR
D. Mul

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Choice Type Award: 1 Penalty: 0.33 CO & Architecture

The method of mapping the consecutive memory blocks to consecutive cache blocks is called .

A. Set associative
B. Associative
C. Direct
D. Indirect

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 02sec Discuss

Q #10 Multiple Choice Type Award: 1 Penalty: 0.33 CO & Architecture

The amount of time required to read a block of data from a disk into memory is composed of seek time, rotational latency, and transfer time. Rotational latency refers to .

A. the time it takes for the platter to make a full rotation
B. the time it takes for the read-write head to move into position over the appropriate track
C. the time it takes for the platter to rotate the correct sector under the head
D. none of the above

Your Answer: Correct Answer: A Not Attempted Time taken: 00min 00sec Discuss

Q #11 Multiple Choice Type Award: 2 Penalty: 0.67 CO & Architecture

Your processor has an 8-line level 1 data cache (direct mapped organization). Suppose that cache lines are 32 bytes (256 bits) and memory addresses are 16 bits, with byte-addressable memory.

Suppose the processor accesses the following data addresses starting with an empty cache:

- 0x0028: 0000 0000 0010 1000
- 0x102A: 0001 0000 0010 1010
- 0x9435: 1001 0100 0011 0101
- 0xEFF4: 1110 1111 1111 0100
- 0xBEEF: 1011 1110 1110 1111
- 0x4359: 0100 0011 0101 1001
- 0x01DE: 0000 0001 1101 1110
- 0x8075: 1000 0000 0111 0101
- 0x9427: 1001 0100 0010 0111

How many hits would there be in the level 1, direct mapped data cache?

A. 2
B. 1
C. 4
D. 3

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

Q #12 Multiple Choice Type Award: 2 Penalty: 0.67 CO & Architecture

Suppose a computer's address size is k bits (using byte addressing), the cache size is S bytes, the block size is B bytes, and the cache is A-way set associative. Assume B is a power of two, so $\mathrm{B} = 2^b$. Find the number of Tag bits needed to implement this cache -

A. $k - \log_2(\mathrm{S} / \mathrm{A})$
B. $k - \log_2(\mathrm{S} / \mathrm{A\ B})$
C. $k - \log_2(\mathrm{S} / \mathrm{B})$
D. $\log_2(\mathrm{S} / \mathrm{A\ B})$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Given a single-level, direct-mapped cache with a $256\ \mathrm{B}$ block size starting in a 'cold' state, i.e., all entries are initially invalid. We store the array 'arr' of integers in main memory with an int-aligned address (i.e. base address divisible by the size of int). The array can be stored anywhere in the main memory block with an integer-aligned address. If we perform sequential accesses to $\mathrm{arr}[0], \mathrm{arr}[1]$, and $\mathrm{arr}[2]$, then what is the probability of getting cache hits for $\mathrm{arr}[1]$ and $\mathrm{arr}[2]$ after the cache miss on $\mathrm{arr}[0]$? Assume the size of int is 4 Bytes.

A. $63/64$
B. $3/4$
C. $31/32$
D. $1/2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider two caches. Both have 16 cache lines, with each cache line being 16 bytes in size. Both caches initially have all lines marked as invalid. The only difference between them is that one is two-way associative, and the other is direct-mapped. Assume that the Main Memory uses 16-bit addresses. What is the length of the shortest access pattern where the two-way associative cache would get a hit, and the direct-mapped cache would get no hits? If you find $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}$ as the shortest access pattern, mark 4 as the answer, as the length is 4, where $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}$ are 16-bit addresses, and these addresses could be different, or any of them could be the same as any other.

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider how the following two C loops would be accessed by a memory system with 32 words and a cache with 4 sets, 2 lines per set, and a block size of 4 words.

```
for (i = 0; i < 32; i++)
    a[i] = i;
sum = 0;
for (i = 0; i < 32; i++)
    sum += a[i];
```

Assume the array is an array of words, only the array is in the memory system, and the first word of the array is at address zero. If the cache is initially empty, how many cache hits would there be for writes and how many cache hits would there be for reads?

A. 32 read hits and 24 write hits.
B. 0 read hits and 0 write hits.
C. 24 read hits and 24 write hits.
D. 0 read hits and 24 write hits.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Select Type | Award: 2 | Penalty: 0 | CO & Architecture

Assume an $N-way$ set-associative cache ($N \geq 2$) and a direct-mapped cache both have the same address size, same data capacity, and same number of index bits. The block sizes & tag bits, however, may be different in the two systems. This means

A. The direct-mapped cache has more tag bits
B. The direct-mapped cache has fewer tag bits
C. The direct-mapped cache has more block offset bits
D. The direct-mapped cache has fewer block offset bits

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Select Type | Award: 2 | Penalty: 0 | CO & Architecture

Given the following series of address references given as word addresses : $2, 3, 11, 16, 21, 13, 64, 48, 19, 11, 3, 22, 4, 27, 6,$ and 11. Assuming a direct-mapped cache with 16 one-word blocks that are initially empty, which of the following is/are NOT the final contents of the cache:

A. 6
B. 19
C. 27
D. 11

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

If an instruction takes 3 cycles for execution, then how many cycles are needed for executing4 instructions of the same type in a sequence using a 3 -stage pipeline?
Assume that there are no interrupts or exceptions while executing them.

A. 12 cycles
B. 6 cycles
C. 9 cycles
D. 4 cycles

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider 16-bit binary representation of decimal number "53". It is stored in a system, which has byte-addressable memory where word size is 2 bytes, beginning at address 100 in big-endian fashion. What would be the content of location addressed by 101, in binary? (At bit-level we use big-endian)

A. 00110101
B. 00000000
C. 10101100
D. 01010011

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

$X(R_i)$ represents an Index addressing mode.

The Effective address of operand is given by $EA = X + [R_i]$

X denotes signed integer and mentioned in the instruction itself. If the contents of the register R_i is always equal to zero then Index addressing mode will implement

A. Immediate addressing mode
B. Absolute addressing mode
C. Indirect addressing mode
D. Register addressing mode

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

In a certain pipelined processor a branch instruction can cause 2 stalls if it is not dealt properly. We can avoid this penalty if a clever compiler can fill the immediate next 2 slots with independent instructions.

Two compilers A and B generate code for this processor.

Compiler A can fill the first delay slot 35% of the time and second delay slot 85% of the time. Compiler B can fill the first delay slot 20% of the time and second delay slot 100% of the time. Assume that 30% of the instructions are branch instructions. What is the $\%$ of the improvement in CPI(Clocks Per Instruction) with compiler A over compiler B?

A. 0
B. 5
C. 10
D. 15

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a processor with a four-stage pipeline $(\text{IF, ID, EX, and WB})$.
The functionality of the stages of this pipeline is as follows :

- IF – Instruction fetch from instruction memory
- RD – Instruction decode and Register read
- EX – Execute-ALU cooperation for data $\&$ address calculation
- WB – Register write back

Consider the following sequence of instructions in which the destination register is the first (leftmost) register. All the instructions use register addressing modes for sources and destination.

```
I1 : SUB   R1,R2,R3
I2 : ADD   R4,R1,R2
I3 : DIV   R5,R1,R4
```

Each instruction in each stage takes one clock cycle. The number of clock cycles required to complete the above sequence of instructions starting from I1, assuming data/operand forwarding is used, is _____

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

A pipeline of 2 stages with a delay of x uanits each is split to n stages. In the new design, each stage has a delay of x/n units. To get the throughput increase of 1700% what would be the n value?

Your Answer: | Correct Answer: 18 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Select Type | Award: 2 | Penalty: 0 | CO & Architecture

Identify the write-read, write-write, and read-write dependencies in the instruction sequence below.

$$\begin{aligned}
&\text{L1: R1} = 100\\
&\text{L2: R1} = \text{R2} + \text{R4}\\
&\text{L3: R2} = \text{R4} - 25\\
&\text{L4: R4} = \text{R1} + \text{R3}\\
&\text{L5: R1} = \text{R1} + 30
\end{aligned}$$

Which of the following is true?

A. Number of write-read(WAR) dependency $= 4$
B. Number of read-write(RAW) dependency $= 4$
C. Number of write-write(WAW) dependency $= 3$
D. Number of write-read(WAR) dependency $= 3$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Select Type | Award: 2 | Penalty: 0 | CO & Architecture

When performing hardware integer arithmetic, we say that overflow has occurred when the mathematically correct result of a computation cannot be represented in the number of bits available for the type being used. Obviously, it is important to be able to detect when an overflow error has occurred.

For the following options, the bit-sequences are 16-bit 2's complement representations of (signed) integer values. For which of the following, an overflow occurs when the given two integers are added?

A. $0111\ 1001\ 1011\ 1011 + 0011\ 1011\ 1110\ 1110$
B. $1111\ 0111\ 0110\ 1001 + 1000\ 0001\ 0110\ 0100$
C. $0001\ 1100\ 0110\ 1111 + 1111\ 0111\ 1110\ 1101$
D. $1100\ 1011\ 1010\ 1101 + 0111\ 1111\ 0010\ 1111$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

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